

21 D

Option C is incorrect since the current is the same in both sections.

Since $R = \frac{\rho l}{A} \Rightarrow \frac{R}{l} = \frac{\rho}{A} \Rightarrow \frac{R}{l} \propto \frac{1}{A}$ (ρ is constant for same material), resistance per unit length of the narrow section is twice that of wide section since the constant current flows through the narrow section.

Since $V \propto R$ for constant I , the resistor with the larger resistance has a larger p.d. across it. Hence, the narrow section (with larger resistance per unit length), has larger p.d. per unit length across it. Hence options A and B are wrong.